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Modular perforation tool

Abstract

A perforation tool for use in a well bore is described herein. The perforation tool comprises a housing; a plurality of frames that fit inside the housing, each frame having a cylindrical shape with a central axis and a plurality of liners, each liner having an axis perpendicular to the central axis, wherein the axes of the liners of each frame are disposed in a plane perpendicular to the central axis, and the frames are axially stackable; an electrical conductor disposed along a central passage of each frame; a plurality of shaped charges secured in the liners of the frames; a bulkhead member disposed in the housing and forming a seal with the housing; and an initiator module disposed in the housing with the bulkhead member between the initiator module and the plurality of frames.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/595,411, filed Nov. 16, 2021, which is a national stage entry of International Application No. PCT/US2020/032879, filed May 14, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/848,856, filed May 16, 2019, and U.S. Provisional Patent Application No. 63/011,082, filed on Apr. 16, 2020. Each of the above applications is incorporated herein by reference in its entirety.

FIELD

(1) Embodiments of the present invention generally relate to well perforation in oil and gas recovery. Specifically, embodiments of a modular perforation tool are described herein.

BACKGROUND

(2) Oil wells are holes drilled into the earth, sometimes under water. The holes may descend essentially straight into the earth, or may advance at an angle, or even sideways. The well provides a conduit for oil and gas, or indeed any flowable resource, to flow out of a geologic formation into the hole and up to the surface. The hole is often lined with a support material to keep the flow path open as fluids flow into the conduit. The support material can be a pipe, a concrete lining, or other support. Oil and gas flow through tiny fissures, cracks, and capillaries in the geologic formation under the influence of a pressure drop between the formation and the hole. As the fluids move through the formation, solids can be entrained in the fluid movement that can be lodged into the cracks and fissures, slowing recovery of fluids from the formation. As fluids are produced from the formation, the pressure of the formation can decline, reducing the pressure drop causing the fluids to flow.

(3) In some cases, flow from the geologic formation can be enhanced by forming conduits from the well wall into the formation. These are openings that extend from the well wall into the formation providing an open pathway substantially immune to plugging with solids to promote free flow of oil and gas into the well from the formation. These conduits are typically formed by inserting a perforation tool into the well to fire explosive charges shaped to propel blast force in a desired pattern toward the well wall, often through a well casing and/or lining. The focused blast force pulverizes and removes a portion of the geology, rock or other geology, at the well wall and a radial distance from the well wall into the formation, sometimes up to 60 inches, to form a conduit. In most cases, more than one such conduit is formed in the well wall at different azimuths and depths depending on the production needs of the well.

(4) A perforation tool, sometimes called a perforating gun, is a complex tool that delivers a firing impulse to one or more explosive charges positioned by the perforating gun to direct the blast from the shaped charge in a desired direction. The explosive charges are contained and held in position by a cage, and a firing circuit is electrically connected to the shaped charges. Transportation of such assemblies is heavily regulated worldwide, due to the propensity to set off an explosion. In most cases, the firing impulse sub-assembly must be transported separately from the explosive sub-assembly, and the tool assembled in the field. Such restrictions complicate delivering perforation tools to drill sites. Additionally, securing the shaped charges in a desired position in the perforating gun is challenging. Often, the shaped charges must be installed at specific positions in the perforating gun, restricting flexibility in applying the blast. A perforating gun is needed in the industry that is transportable and has more flexibility of operation.

SUMMARY

(5) Embodiments described herein provide an apparatus, comprising a housing; a plurality of frames that fit inside the housing, each frame having a cylindrical shape with a central axis and a plurality of liners, each liner having an axis perpendicular to the central axis, wherein the axes of the liners of each frame are disposed in a plane perpendicular to the central axis, and the frames are axially stackable; an electrical conductor disposed along a central passage of each frame; a plurality of shaped charges secured in the liners of the frames; a bulkhead member disposed in the housing and forming a seal with the housing; and an initiator module disposed in the housing with the bulkhead member between the initiator module and the plurality of frames

(6) Other embodiments described herein provide a perforation tool, comprising a cylindrical housing having a plurality of perforation grooves formed therein; a plurality of charge frames that fit inside the housing, each charge frame having a cylindrical shape with a central axis and a plurality of liners positioned adjacent to one of the perforation grooves, each liner having an axis perpendicular to the central axis, wherein the axes of the liners of each charge frame are disposed in a plane perpendicular to the central axis, each liner has a retention member that extends around a circumference of the liner, and the charge frames are axially stackable; a plurality of shaped charges secured in the liners of the charge frames; a bulkhead member disposed in the housing and

forming a seal with the housing; and an initiator module disposed in the housing with the bulkhead member between the initiator module and the plurality of charge frames.

(7) Other embodiments described herein provide a perforation tool, comprising a cylindrical housing; a first perforation assembly disposed inside the housing, that first perforation assembly comprising a first frame for shaped charges, an initiator module, and a bulkhead member between the frame and the initiator module; and a second perforation assembly disposed inside the housing, the second perforation assembly comprising a second frame for shaped charges electrically coupled to the initiator module of the first perforation assembly by an electrical conductor disposed through a central passage of the second frame.

(8) Other embodiments described herein provide a frame for housing shaped charges, comprising a body having a central axis, with a passage through the body along the central axis and a plurality of openings oriented transverse to the central axis, each opening comprising a wall having circular cross-section and diameter that declines from a first end of the opening to a second end of the opening, the first end of the opening having a circular rim and the second end of the opening having a port that fluidly couples the opening with the passage, each opening having a notch in the wall, the notch having a first side wall and a second side wall opposite the first side wall, with a restraint extending from the first side wall across the notch toward the second side wall.

(9) Other embodiments described herein provide a frame for housing shaped charges, comprising a body having a central axis, with a passage through the body along the central axis and a plurality of openings oriented transverse to the central axis, each opening comprising a wall having circular cross-section and diameter that declines from a first end of the opening to a second end of the opening, the first end of the opening having a circular rim and the second end of the opening having a port that fluidly couples the opening with the passage, each opening having two rectangular notches in the wall on opposite sides, the two notches aligned along the central axis, with a restraint extending across each notch along the circumference of the opening.

(10) Other embodiments described herein provide a frame for housing shaped charges, comprising a body having a central axis, with a passage through the body along the central axis and a plurality of openings oriented transverse to the central axis, each opening comprising a wall having circular cross-section and diameter that declines from a first end of the opening to a second end of the opening, the first end of the opening having a circular rim and the second end of the opening having a port that fluidly couples the opening with the passage, each opening having two rectangular notches in the wall on opposite sides, the two notches aligned along the central axis, with a restraint extending across each notch along the circumference of the opening, the body having a first end with a plurality of first alignment features and a second end opposite the first end with a plurality of second alignment features that match the first alignment features.

(11) Other embodiments described herein provide an initiator module, comprising a first member having a contact ring with a tapered surface and a central opening; and a second member comprising a circuit plate coupled to the second member in a transverse orientation by a plurality of clips; and a central plug holding a detonator that protrudes through the central opening of the first member.

(12) Other embodiments described herein provide an initiator module, comprising a first member having a contact ring at a first end of the initiator module, the contact ring having a tapered surface and a central opening; and a second member comprising a circuit plate coupled to the second member by a plurality of clips; and a central plug holding a detonator that protrudes through the central opening of the first member, wherein the initiator module has a cylindrical shape with a central axis, the circuit plate is oriented along a plane perpendicular to the central axis, the initiator module has a second end opposite from the first end, and the initiator module further comprises an electrical connector that protrudes from the second end.

(13) Other embodiments described herein provide an initiator module, comprising a first member having a contact ring at a first end of the initiator module, the contact ring having a tapered surface

and a central socket defining a central opening; and a second member comprising a circuit plate coupled to the second member by a plurality of clips; and a central plug holding a detonator that fits within the central socket of the first member, wherein the initiator module has a cylindrical shape with a central axis, the circuit plate is oriented along a plane perpendicular to the central axis, the initiator module has a second end opposite from the first end, and the initiator module further comprises a cylindrical electrical connector that protrudes from the second end.

(14) Other embodiments described herein provide a bulkhead member for a perforation tool, the bulkhead member comprising a cylindrical body made from a structurally strong material the cylindrical body having a first end, a second end opposite the first end, and a central passage formed along an axis thereof from the first end to the second end; and an electrical conductor disposed in the passage along the axis and having a first end with a rounded prong that extends into a socket formed in the first end of the cylindrical body, and a second end that has an electrical coupling disposed at the second end of the cylindrical body.

(15) Other embodiments described herein provide a bulkhead member for a perforation tool, the bulkhead member comprising a cylindrical body made from a structurally strong material the cylindrical body having a first end, a second end opposite the first end, and a central passage formed along an axis thereof from the first end to the second end; and an electrical conductor disposed in the passage along the axis and having a first end with a rounded prong that extends into a socket formed in the first end of the cylindrical body, and a second end that has an electrical coupling disposed at the second end of the cylindrical body, wherein the first end of the electrical conductor is electrically connectable to an initiator module for the perforation tool and the second end of the electrical conductor is electrically connectable to an initiator module for the perforation tool and to a charge module for the perforation tool.

(16) Other embodiments described herein provide a bulkhead member for a perforation tool, the bulkhead member comprising a cylindrical body made from a structurally strong material the cylindrical body having a first end, a second end opposite the first end, and a central passage formed along an axis thereof from the first end to the second end; an electrical conductor disposed in the passage along the axis and having a first end with a rounded prong that extends into a socket formed in the first end of the cylindrical body, and a second end that has an electrical coupling protruding from the second end of the cylindrical body, wherein the first end of the electrical conductor is electrically connectable to an initiator module for the perforation tool and the second end of the electrical conductor is electrically connectable to an initiator module for the perforation tool and to a charge module for the perforation tool; and an insulating member disposed in the central passage between the cylindrical body and the electrical conductor.

(17) Other embodiments described herein provide a method of downhole processing, comprising positioning a perforating charge in a perforating tool; positioning a booster charge adjacent to the perforating charge; deploying the perforating tool in a well; and activating the perforating charge by directly transferring ballistic energy from the booster charge to the perforating charge.

(18) Other embodiments described herein provide a method of downhole processing, comprising positioning a plurality of perforating charges in a perforating tool; positioning a plurality of booster charges adjacent to the perforating charges, each booster charge adjacent to a corresponding perforating charge; deploying the perforating tool in a well; activating a first booster charge of the plurality of booster charges; activating each booster charge by direct transfer of ballistic energy from a neighboring booster charge; and activating each perforating charge by direct transfer of ballistic energy from the corresponding booster charge.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

(2) FIG. 1A is a side view of a perforation tool according to one embodiment.

(3) FIG. 1B is a detailed view of a portion of the perforation tool of FIG. 1A.

(4) FIG. 1C is a cross-sectional view of a portion of the perforation tool of FIG. 1A.

(5) FIG. 1D is a more detailed cross-sectional view of a portion of the perforation tool of FIG. 1A.

(6) FIG. 2A is a cross-sectional view of a perforation tool in a transportation configuration according to another embodiment.

(7) FIG. 2B is a detail view of a portion of a perforation tool, according to one embodiment.

(8) FIG. 2C is a cross-sectional view of an initiator module according to another embodiment.

(9) FIG. 3A is an isometric view of a charge frame according to another embodiment.

(10) FIG. 3B is a cross-sectional view of a portion of the charge frame of FIG. 3A.

(11) FIG. 3C is a different isometric view of the frame of FIG. 3A.

(12) FIG. 3D is an isometric view of a frame according to another embodiment.

(13) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(14) The modular perforation tools described herein are configured to be transportable in a single parcel, and to be scalable to any desired degree in the field. The tools described herein are also feature configurable discharge directionality and spacing. FIG. 1A is a side view of a perforation tool **100** according to one embodiment. The perforation tool **100** is configured to integrate into a tool for use in a well bore. The perforation tool **100** has a first end **102** and a second end **104** opposite from the first end **102**. Each end is configured to attach to a drill string, by threaded connection, bolting, or other attachment mode. The apparatus shown in FIG. 1A displays an outer housing **106**, which is in three identical segments **106A**, **106B**, and **106C**, which are fit together at joints **107A** and **107B**. Each of the segments **106A**, **106B**, and **106C** has an initiator end **108** and a bulkhead end **110**. The initiator end **108** can house an initiator and the bulkhead end **110** can house a bulkhead, each of which is further described below. Each segment **106A**, **106B**, and **106C** also has a charge region **112**, generally between the initiator end **108** and the bulkhead **110**, which houses shaped charges (e.g., perforating charges) that perform the perforation of the well bore when activated. The outer housing **106** of the perforation tool has a generally cylindrical shape with circular cross-section at most locations along its length, and a central cylindrical axis **103**. Thus, the initiator end **108** of each segment **106A**, **106B**, and **106C** encloses a cylindrical interior in which a cylindrical object can be disposed. Likewise, the bulkhead end **110** of each segment **106A**, **106B**, and **106C** encloses a cylindrical interior, and the charge region **112** also encloses a cylindrical interior.

(15) FIG. 1B is a detail view of a portion of the perforation tool **100** of FIG. 1A. This view shows one of the segments **106A**, **106B**, **106C**, of the outer housing **106**. The outer housing **106** has a wall thickness that varies according to the function of the particular location in the outer housing **106**, some parts thicker and some parts thinner. In particular, each charge region **112** features two perforation grooves **114**, a first perforation groove **114A** and a second perforation groove **114B**, which are thin portions of the outer housing **106** configured to admit a discharge impulse of gas

from one or more shaped charges located in the interior of the charge region **112** adjacent to each perforation groove **114**. Each perforation groove **114** is a band formed around the circumference of the outer housing **106**, generally with wall thickness that is less than a wall thickness at another location in the housing. The band proceeds around the circumference of the outer housing **106** in an azimuthal direction, generally along a plane transverse to the central axis. Each perforation groove **114** has a flat central portion **116** and a sloped wall **118** on either side of the central portion **116** that connects the central portion **116** to an outer wall **120** of the outer housing **106**. The first perforation groove **114A** has a first central portion **116A**, a first sloped wall **118A** and a second sloped wall **118B**, opposite from the first sloped wall **118A**. The second perforation groove **114B** has a second central portion **116B**, a third sloped wall **118C** and a fourth sloped wall **118D**, opposite from the third sloped wall **118C**. The sloped walls **118** may be flat or curved, and may form an angle with the central portion of 20-80 degrees. In this case, the sloped walls **118** all form an angle of 45 degrees with the respective central portions **116**.

(16) The central portions **116** of the perforation grooves **114** have the same width along the axial direction of the perforation tool **100**. The width of the central portions **116** is selected to provide a window for discharge gas penetration of the central portion tailored to result in a discharge impulse, upon charge activation, that travels from the central portion into the well bore wall a desired distance with a desired dispersion profile. If the internal diameter of the outer housing **106** at the central portion **116** is D and the width of the central portion **116** is W , a ratio D/W is general from about 1 to about 100, such as from about 2 to about 10, for example about 3.

(17) Here, there are two perforation grooves **114**, but the outer housing **106** can have any desired number of perforation grooves **114**, as will be further explained below due to the modularity of the perforation tool **100**. A minimum spacing between the perforation grooves **114** is generally set by interior structures of the perforation tool **100**. Here, the spacing between the perforation grooves **114** is about the same as the width of the central portions **116**. Each segment **106A**, **106B**, and **106C** of the outer housing **106**, in this case, has a bulkhead region **122** extending from the bulkhead end **110** of the segment to the first sloped wall **118A** of the first perforation groove **114A**, a first central portion **116A** extending from the first sloped wall **118A** to a second sloped wall **118B** of the first perforation groove **114A**, opposite the first central portion from the first sloped wall **118A**, a spacing region **126** extending from the second sloped wall **118B** of the first perforation groove **114A** to the third sloped wall **118C** of the second perforation groove **114B**, a second central portion **116B** extending from the third sloped wall **118C** to the fourth sloped wall **118D**, opposite the second central portion **116B** from the third sloped wall **118C**, and an initiator region **128** extending from the fourth sloped wall **118D** to the initiator end **108** of the segment.

(18) FIG. 1C is a cross-sectional view of the perforation tool **100** of FIG. 1A. The outer housing **106** of the perforation tool **100**, in this case, has three segments **106A**, **106B**, and **106C**, as mentioned above. The first and third segments **106A** and **106C** are shown here largely empty in order to present the internal wall structure of the outer housing **106** clearly. The second segment **106B** mainly features the internal structures of the perforation tool **100**.

(19) The perforation tool **100** includes an initiator module **130**, a plurality of charge modules **132**, in this case two charge modules **132** matched with the number of perforation grooves **114** in the outer housing **106**, and a bulkhead member **134**. Two bulkhead members **134** are shown in the apparatus of FIG. 1C. The charge modules **132** are arranged between a bulkhead member **134** and the initiator module **130**. The initiator module **130** is a cylindrical module with an outer radius substantially the same as an inner radius of the outer housing **106** at the initiator region **128**.

(20) The wall of the outer housing **106** at the initiator region **128** has a first section **136**, a second section **138**, and a third section **140**. The first section **136** starts at the initiator end **108** of the outer housing **106** and extends to the second section **138**. The second section **138** is between the first section **136** and the third section **140**. The inner diameter of the outer housing **106** tapers in the first section **136** from a first diameter at an end of the first section **136** to a second diameter at the

second section **138**. The first diameter is less than the second diameter. The tapered portion of the inner wall of the outer housing **106** provides a tapered surface **139** for engaging with the initiator module **130**.

(21) The initiator module **130** comprises a circuit plate **142** disposed in a container **144**. The container **144** comprises a first member **146** and a second member **148**. The first and second members **146** and **148** can be separated to extract the circuit plate **142** from the container **144**. The initiator module **130** has a generally cylindrical shape with a first end **152** and a second end **185**. The first member **146** has a contact ring **150** at the first end **152**. The contact ring **150** has a tapered surface **154** that matches the tapered wall of the outer housing **106** in the first section **136** of the initiator region **128** thereof. The tapered surface **154** of the contact ring **150** has an outer radius that matches the inner radius of the wall of the outer housing **106** in the first section **136**, the outer radius linearly declining toward the first end **152** such that the tapered surface **154** contacts the tapered wall of the outer housing **106** in the first section **136**. The outer radius can generally have any declining profile that matches with the inner radius of the wall of the outer housing **106**, such as curved or angled. It should be noted that, instead of providing tapered surfaces for engagement of the contact ring **150** with the wall of the outer housing **106**, the wall of the outer housing **106** could have a shelf extending from a right cylindrical wall, and the first member could have a contact ring, or contact portion, with a straight surface to engage with the wall and an end to contact the shelf.

(22) The first member **146** of the container **144** has a central opening **156** that accommodates a detonator **158** that is disposed in the initiator module **130**, and when so disposed, protrudes through the central opening **156**. The second member **148** includes a central plug **160** that is aligned with the central opening **156** and holds the detonator **158** in position. The second member **148** also includes a plurality of clips **162** that hold the circuit plate **142** in a position transverse to the central axis **103**. The central plug **160** can be formed integrally with the second member **148**, or may be a separate member that engages with the rest of the second member **148** and with the detonator **158** and the first member **146**.

(23) The first member **146** has a central socket **155** that defines the central opening **156** and extends along an axis of the initiator module **130**. The central plug **160** fits within the central socket **155** to assemble the first member **146** and the second member **148**. The central plug **160** and central socket **155** may be electrically conductive to provide electrical coupling from the initiator module **130** to the charge module **132**, as further described below. Alternately, one or both of the central plug **160** and the central socket **155** may be non-conductive materials, for example polymer, with conductive members such as wires or rods disposed therein, for example in grooves in the surface of either or both of the central plug **160** and the central socket **155**, or disposed within the body of either or both of the central plug **160** and the central socket **155**.

(24) The central socket **155** has a nub **145** that extends slightly beyond the end of the detonator **158** and encloses the detonator **158**. The central socket **155** has three sections, a first section that engages with the central plug **160**, a second section that holds the detonator **158**, and a third section defined by the nub **145**. Each section is cylindrical in profile. The first section has an inner diameter larger than an outer diameter of the detonator **158** to accommodate the central plug **160**. The second section has an inner diameter substantially the same as the outer diameter of the detonator **158** in order to hold the detonator **158**. Here, the first and second sections have the same outer diameter, but the outer diameter of the two sections could be different. The third section, defined by the nub **145**, has an inner diameter substantially the same as the outer diameter of the detonator **158**, and an outer diameter less than the outer diameter of the first and second sections. The nub **145** has a hollow end to provide fluid coupling through the end of the nub **145** for ballistic coupling to a charge module **132**. The nub **145** fits within an electrical coupling of a charge module **132**, as further described below, to provide electrical coupling to the charge module **132**.

(25) Here, no wires are shown for connecting the detonator **158** to electricity. The detonator **158**

may receive electrical energy by contact with the central plug **160**, for example if the detonator **158** has electrical contacts built in. Alternately, the detonator **158** may have wires. In this case, the wires are disposed through a central opening **161** of the second member **148** to make electrical connection with contacts of the circuit plate **142**.

(26) FIG. 1D is a cross-sectional view of the charge modules **132** of the perforation tool **100**. Each charge module **132** includes a frame **164** housing a plurality of shaped charges **166**, each shaped charge disposed in a liner **168** that engages with the frame **164**. Each frame **164** has a cylindrical shape with a central axis **165** aligned with the central axis **103** of the perforation tool **100** when the frames **164** are installed in the perforation tool **100**. The liners **168** are disposed in openings **170** of the frame **164**, the openings **170** arranged along a plane **172** perpendicular to the central axis **103**. The plane **172** is here represented as a dotted line, since the view of FIG. 1D shows the plane edge-on. Each frame **164**, in this case, has three such openings **170** spaced 120° apart around the azimuth of the frame **164**. The liners **168** are cup-shaped to fit the shape of the charges. The frames **164** are positioned in the interior of the outer housing **106** such that the openings **170**, liners **168**, and charges are positioned adjacent to a perforation groove **114** of the outer housing **106**. Depending on the size of the charges to be used, the frames **164** can be configured to hold any convenient number of charges in openings spaced uniformly around the azimuth of the frame. Thus, two, three, four, five, or more charges can be positioned in a frame similar to these frames **164** by configuring the openings of the frames.

(27) The frames **164** are made of a polymer material such as polypropylene, polyurethane, or another strong polymer material. The frames **164** can be made by molding, sculpting (for example laser scribing), or 3D printing. As shown in FIGS. 1B-1D, the frames **164** are stackable such that any number of frames **164** can be disposed in a perforation tool. Usually the number of frames **164** disposed in the perforation tool will be consistent with the number of perforation grooves **114** in the outer housing **106**, but a greater or lesser number of frames **164** can, in principle, be disposed in the outer housing **106**. The frames **164** are also rotatable within the outer housing **106** to orient the discharge direction in any desired way. The perforation grooves **114**, which proceed completely around the housing **106**, generally allow orienting a shaped charge in any azimuthal direction. The perforation grooves **114** are equally susceptible to penetration at every azimuth by the shaped charges located within the frames **164** adjacent to the perforation grooves **114**. Rotatability of the frames **164** supports pointing the shaped charges held by the frames **164** in any direction within the perforation tool, without limitation. In this way, the housing **106** of the perforation tool **100** does not need to be positioned or oriented according to a rotation angle to allow jets from the shaped charges to penetrate in a desired direction.

(28) Each of the frames **164** has a central passage **174** axially oriented in the center of the frame **164**. An electrical conductor **176** is disposed through the central passage **174** of each frame **164**. In this case, each frame **164** has an electrical conductor **176** disposed in the central passage **174** thereof. Each electrical conductor **176** contacts an inner wall **178** of the passage **174**, but a space can be provided between the electrical conductor **176** and the inner wall **178** in some cases. Such space can be a continuous annular space, or may be discontinuous, if desired.

(29) Each electrical conductor **176** has a first end **180** and a second end **182** opposite the first end **180**. The first end **180** has a coupling **184**. The coupling **184** has a first diameter and the rest of the electrical conductor **176** has a second diameter. In this case, the first diameter is greater than the second diameter. The coupling can receive a connection portion of another module, such as the initiator module **130** or another frame **164**. The second end **182** of each electrical conductor **176** protrudes beyond the frame **164**, providing a connection that can connect the frame **164** to another frame **164**, an initiator module **130**, or a bulkhead member **134**. When the charge modules **132** and initiator module **130** are installed in the perforation tool **100**, the coupling **184** of a frame **164** receives the portion of the detonator **158** that protrudes through the first member **146** of the initiator module **130**. Here, the central passage **174** has a first section **186** with a first diameter and

a second section **188** with a second diameter. The first diameter, in this case, is greater than the second diameter. Here, the second diameter of the second section **188** of the central passage **174** is substantially the same as the outer diameter of the electrical conductor **176**, while the first diameter of the central passage **174** is substantially the same as an outer diameter of the coupling **184**. Thus, an exterior surface of the coupling **184** contacts the first section **186** of the central passage **174** and the rest of the electrical conductor **176** contacts the second section **188** of the central passage **174** such that there is essentially no space between the electrical conductor **176** and the central passage **174**. The coupling **184** in this case has a tapered outer surface where the coupling **184** joins the rest of the electrical conductor **176**. The tapered outer surface of the coupling contacts a similarly tapered portion of the inner wall **178** of the central passage **174** that joins the first section **186** and second section **188** of the central passage **174**. As noted above, this construction allows for space to be provided between the electrical conductor **176** and the second section **188** of the central passage **174**, if desired.

(30) A capsule **190** is positioned within the electrical conductor **176** for each frame **164** adjacent to the openings **170** of the frame, and adjacent to the shaped charges **166**. The capsule **190** is made of a combustible material that provides an ignition source for the shaped charges. Unlike conventional boosters used in downhole tools, the capsule **190** has no direct physical connection to an energy source such as a detonator cord. The capsule **190** is merely a shaped volume of combustible material, perhaps contained in a wrapping, that fits within the electrical conductor **176**. The capsule **190** is, thus, wireless. The capsule **190** is slipped into the electrical conductor **176** before the frame **164** is disposed in the container **144**. Here, the electrical conductor **176** has a ridge **192** at one end of the electrical conductor **176** to aid in dependable placement of the capsule **190** with respect to the shaped charges. In this case, the ridge **192** is located at the second end **182** of the electrical conductor **176**, but the ridge **192** could be positioned near the first end **180** of the electrical conductor **176**.

(31) To hold the capsule **190** in place inside the electrical conductor **176**, a restraint **194** is positioned inside the electrical conductor **176** at a desired location. The restraint **194** can be a removable member, such as a snap ring, or a retractable member, which can extend around the entire circumference of the electrical conductor **176** or partway around the circumference. In any event, the restraint **194** can be removed or repositioned to allow placement of the capsule **190** between the restraint **194** and the ridge **192**. In FIG. 1D, the capsule **190** is smaller than the space between the ridge **192** and the restraint **194**.

(32) In other embodiments, the capsule **190** may be sized to extend substantially from the first end **180** to the second end **182** of the electrical conductor **176**. In such cases, the capsule **190** can be held in place between the ridge **192** of a first electrical conductor **176** and the ridge **192** of a second electrical conductor **196**. Thus, referring to FIG. 1D, a capsule **190** might extend from a ridge **192** at the second end **182** of a first electrical conductor **176** to the first end **180** of the first electrical conductor **176**, and may be held in place between the ridge **192** of the first electrical conductor **176** and the second end **182** of a second electrical conductor **176** disposed against the first end **180** of the first electrical conductor **176**. Use of a separate restraint **194**, as shown in FIG. 1D, allows for use of smaller capsules **190**.

(33) An opening in the electrical conductor **176**, adjacent to a similar opening in the inner wall **178** of the central passage **174**, provides fluid coupling from the interior of the electrical conductor **176** through the frame **164** to the shaped charges **166**. Each electrical conductor **176** provides fluid coupling along the length thereof from the detonator **158** to the capsules **190**. The electrical conductor **176** thus provides electrical and ballistic coupling from the initiator module **130** to the shaped charges **166**. When the circuit plate **142** provides an electrical impulse, the electrical impulse is transmitted to the detonator **158** and to a first electrical conductor **176** of a first frame **164** adjacent to the initiator module **130**. The electrical impulse is transmitted by the first electrical conductor **176** through the first frame **164** to the capsule **190** of the first electrical conductor **176**.

The detonator **158** combusts upon application of the electrical discharge from the circuit plate, and the ballistic discharge from the detonator **158** travels down the first electrical conductor **176** to the first capsule **190**, which combusts. Discharge from the first capsule **190** discharges the shaped charges **166** adjacent to the first capsule **190**. Discharge from the first capsule **190** also travels down the first electrical conductor **176** to the second end thereof, along with the electrical impulse from the firing of the circuit plate **142**. The second end of the first electrical conductor is mated with the coupling **184** of a second electrical conductor **176** of a second frame **164**, providing fluid and electrical communication from the first electrical conductor **176** to the second electrical conductor **176**. The ballistic discharge, and electrical impulse, travel from the first electrical conductor **176** of the second electrical conductor **176**, to the second capsule **190**, which discharges and ignites the shaped charges adjacent to the second capsule **190**. The discharge from the shaped charges **166** is configured to pierce the perforation grooves **114** of the outer housing **106** with enough energy to continue to the well bore and through the bore wall into the formation. In this way, any number of shaped charges **166** can be discharged using the structure of the perforation tool **100** shown in FIGS. **1A-1D**.

(34) The structures described above allow for activation of perforating charges in a perforating tool without use of wires to transfer energy. The detonator is activated in the initiator module by electrical stimulation, but then a direct transfer of ballistic energy from the detonator activates the first capsule, which in this case is a booster charge. Ballistic energy transfer from activation of the first booster charge activates a first perforation charge adjacent to the first booster charge. Ballistic energy transfer from activation of the first booster charge also activates a second booster charge adjacent to the first booster charge. Ballistic energy transfer from the second booster charge, in turn, activates a second perforation charge adjacent to the second booster charge. In this way, an entire perforation gun made up of a number of perforating charges can be activated essentially wirelessly. The only wired connection in such a structure is from the firing circuit of the circuit plate to the detonator in the initiator module. In the event a wireless initiator module is used with the structures above, the entire perforating tool is wireless.

(35) The circuit plate **142** is also electrically connected to the electrical conductor **176**, which provides electrical power through the frames **164**. The initiator module **130** includes a connector **141** with a central passage to receive the detonator **158** therein. The connector **141** protrudes from the initiator module **130** like the detonator **158** and electrically couples to the coupling **184** of the electrical conductor **176**. The plug **160** inserts into the connector **141**, and the detonator **158** held in the plug **160** passes through the connector **141** and protrudes into the electrical conductor **176** when the initiator module **130** is coupled to the charge module **132**.

(36) The bulkhead member **134** is a dense structurally strong object made, for example, of steel. The bulkhead member **134** has a cylindrical body **147** in this case, but in general is shaped to match the cross-sectional profile of the outer housing **106**. The bulkhead member **134** is disposed in the outer housing **106** in the bulkhead region **122** thereof. As shown in FIG. **1C**, the bulkhead member **134** is inserted into the bulkhead region **122** of the second segment **106B** of the outer housing **106** such that a first end **167** of the bulkhead member **134** couples with the protrusion of the second end **182** of the electrical conductor **176** by a bulkhead conductor **177** inserted through a central passage **175** of the bulkhead member **134** that extends from the first end **167** to a second end **163** of the cylindrical body **147** of the bulkhead member **134**. As noted above, the cylindrical body **147** of the bulkhead member **134** is made of a structurally strong material, which may be electrically conductive or non-conductive. The bulkhead conductor **177** has a first end **173** and a second end **171** opposite from the first end **173**. The first end **173** has a coupling **169** that protrudes at the first end **167** of the bulkhead member **134** and receives an electrical conductor from another module, such as the second end **182** of the electrical conductor **176** of the frame **164** or the connector **141** of the initiator module **130** that protrudes from the initiator module **130**. The second end **171** of the conductor **177** is a prong-like structure with a rounded tip extending into a socket **179** at the second

end **163** of the bulkhead member **134**. The socket **179** is configured to couple only to the second member **148** of the initiator module **130** such that the circuit plate **142** installed in the second member **148** is substantially protected from detonation of the shaped charges in the charge module **132**. In FIG. **1C**, two bulkhead members **134** are installed at either end of the assembly to isolate the discharge of the charge modules **132**.

(37) The bulkhead member **134** is inserted into the housing **106** and abuts a ledge **159** of the outer housing **106** that marks a boundary between the bulkhead region **122** of the outer housing **106** and the charge region **112**. The bulkhead member **134** is a cylindrical object, in this case, but is generally shaped according to the profile of the housing **106**. The bulkhead member **134** has a retention shelf **157** that engages with an edge **143** of the first segment **106A** of the outer housing **106**, which modularly engages with the second segment **106B**, as described above. The edge **143** of the outer housing **106** securely captures the bulkhead member **134** between the edge **143** of one housing segment and the ledge **159** of another housing segment, holding the bulkhead member **134** securely in the bulkhead region **122** of the housing segment.

(38) A first seal is disposed in a first seal groove **153** of the bulkhead member **134** near the first end **167** thereof and between the bulkhead member **134** and the second housing segment **106B**. A second seal is disposed in a second seal groove **151** disposed between the first housing segment **106A**, near the edge **143**, and the bulkhead member **134**. The first seal groove **153** is disposed in a side wall of the bulkhead member **134** that extends from the retention shelf **157** to the first end **167** of the bulkhead member **134** and has a first diameter. The second seal groove **151** is disposed in a side wall of the bulkhead member **134** that extends from the retention shelf **157** toward the second end **163** of the bulkhead member **134** and has a second diameter. The retention shelf **157** is thus located between the first seal groove **153** and the second seal groove **151**, and the first and second seal grooves **153** and **151** provide sealing between the bulkhead member **134** and respective first and second housing segments **106A** and **106B**. The first diameter of the side wall of the bulkhead member **134**, between the first end **167** and the retention shelf **157**, is larger than the second diameter of the side wall, between the retention shelf **157** and the second end **163**, such that the retention shelf **157** can provide secure positioning of the bulkhead member **134** in the outer housing **106**.

(39) The bulkhead member **134** has a recess **149** formed in the first end **167** thereof. The central passage **175** of the bulkhead member **134** is formed at a central area of the recess **149**. The recess **149** facilitates inserting and removing the conductor **177** in the central passage. The second end **163** of the bulkhead member **134** is designed, overall, to abut the initiator module **130**, while the first end **167** of the bulkhead member **134** is designed to interface with a charge module **132**. Here, the first end **167** of the bulkhead member **134** does not abut the charge module **132**, since the conductors of the two components protrude and engage without the modules abutting. By reducing the distance of the conductors of the two modules protrude, the charge module **132** could be disposed close enough to the bulkhead member **134** to abut the first end **167** of the bulkhead member **134**. It should be noted that the bulkhead member **134** is radially symmetric, so rotational orientation of the bulkhead member **134** in the outer housing **106** does not affect its operation.

(40) FIG. **2A** is a cross-sectional view of a perforation tool **200** in a transportation configuration according to another embodiment. The perforation tool **200** packages an initiator module **202** with a charge module **204** in one package for shipment. The initiator module **202** is like the initiator module **130** of FIGS. **1C-1D**, with a two-piece container **206** housing a circuit plate **208** and a detonator **210**. The detonator **210** is located at a first end **212** of the initiator module **202**, and the initiator module **202** is coupled to a bulkhead member **214** at a second end **216** of the initiator module **202**. The bulkhead member **214** is between the initiator module **202** and the charge module **204** in the perforation tool **200**, with the detonator **210** spaced apart from the charge module **204**, such that the perforation tool **200** can be shipped with detonator and charges in the same package, without the possibility of unintended ignition. An electrical connector **217** protrudes from the

second end **216** of the initiator module **202** and engages with a socket **219** of the bulkhead member **214** with a bulkhead conductor **221** positioned in the socket **219** and extending through the bulkhead member **214**. Here, the electrical connector **217** is cylindrical, but in other embodiments the electrical connector **217** may be prongs or brackets that engage with one or more similarly shaped sockets **219**. The electrical connector **217** provides electrical continuity between the bulkhead member **214** and the initiator module **202**. The bulkhead member **214** may be electrically conductive. In such cases, an insulating member **223** is disposed through the bulkhead member **214** between the cylindrical body **147** and the bulkhead conductor **221**, thus preventing short circuiting through the bulkhead member **214**. The insulating member **223** is disposed in the central passage **175** of the cylindrical body **147** (FIG. 1C), and in this case continuously lines the central passage **175**. Thus, in this case, the insulating member **223** is a liner. In other cases, the insulating member **223** may be a spacer instead of, or in addition to, being a liner. In such cases, the insulating member **223** may be discontinuous or may have holes or openings that provide space between the cylindrical body **147** and the bulkhead conductor **221**. In the event the bulkhead member **214** is not electrically conductive, the insulating member **223** can be omitted.

(41) The electrical connector **217** could be a separate member from the initiator module **202**. In one case, the electrical connector **217** could be an electrical contact band that fits within the shaped socket **219**. FIG. 2B is a detail view of a portion of a perforation tool, according to one embodiment. FIG. 2B shows an embodiment in which an electrical contact band **217A** is disposed in the shaped socket **219** to provide electrical connection from the circuit plate **208** to the bulkhead conductor **221**. The electrical contact band **217A** is a generally cylindrical electrically conductive member that fits within the shaped socket **219** and provides electrical connection under a 360 degree rotation of members in electrical communication therewith. The electrical contact band **217A** has an open middle so that the bulkhead conductor **221** can be inserted into the middle of the electrical contact band **217A**. The electrical contact band **217A**, in this case, specifically provides electrical connection with the bulkhead conductor **221** under any rotation of the bulkhead conductor **221**.

(42) In FIG. 2A, the circuit plate **208** is shown electrically connected to the connector **217** by a wire **240**, optionally terminating in a contact pad **242** that connects with the connector **217**. In FIG. 2B, an optional pogo pin **209** can be provided extending from the circuit plate **208** to provide a robust electrical connection from the circuit plate **208** to the connector **217** of FIG. 2A or the electrical contact band **217A** of FIG. 2B. In FIG. 2B, electrical connection from the circuit plate **208** to the electrical contact band **217A** is made at the ring end of the electrical contact band **217A** facing the circuit plate **208** by providing a cylindrical pogo pin **209** with a tapered tip and a diameter larger than an inner diameter of the ring end of the electrical contact band **217A**. Upon deploying the initiator module **202** with the charge module **204**, the tapered tip of the pogo pin **209** nests in the ring end of the electrical contact band **217A**, and the spring-loaded pogo pin **209** depresses to provide an engaging force between the pogo pin **209** and the ring end of the electrical contact band **217A**, thus creating a robust electrical connection between the circuit plate **208** and the bulkhead conductor **221**. While the pogo pin **209** of FIG. 2B is shown engaging with the electrical contact band **217A** in the center of the ring end of the electrical contact band **217A**, the electrical contact could be made using a pogo pin such as the pogo pin **209** at any convenient location between the circuit plate **208** and the bulkhead conductor **221** using wires and contact pads or recesses in any convenient configuration. An RCA connection can also be used in place of the pogo pin **209**.

(43) Here, the electrical contact band **217A** is shown as a cylindrical member. The electrical contact band **217A** may have solid cylindrical walls, or may have vertically slotted walls in some embodiments. In other embodiments, the electrical contact band **217A** may have a generally cylindrical shape with a first diameter at the ends of the electrical contact band and a second diameter near a middle of the electrical contact band, where the second diameter is less than the

first diameter, for example a cylindrical shape with a slight narrowing in the middle. Use of such a shape along with vertical slotting can provide a flexible and secure connection with the electrical contact band **217A**.

(44) There are two charge modules **204** in the perforation tool **200**, but as described elsewhere herein, more or fewer charge modules **204** can be used. The charge modules **204**, bulkhead member **214**, and initiator module **202** are configured with an outer housing **213**, which has a bulkhead region **218**, a charge region **220**, and an initiator region **222**, similar to the outer housing **106**, and segments thereof **106A**, **106B**, and **106C**, described in connection with FIGS. **1A-1D**. To prepare the perforation tool **200** for operation, an initiator module is inserted into the interior of the outer housing **213** in the initiator region **222**, coupling the detonator **210** of the initiator module with the charge module **204**. The initiator module **202** can be disconnected from the bulkhead member **214** and inserted into the initiator region **222**. Alternately, an initiator module of another unit like the perforation tool **200** can be coupled at the initiator region **222**, coupling two charge regions together in an extended perforation tool. The modularity and safe shipping configuration of the perforation tool **200** allows a plurality of such tools to be shipped to a field location and assembled with ease into an operational configuration.

(45) FIG. **2C** is a cross-sectional view of an initiator module **250** according to another embodiment. The initiator module **250** is adapted for use with a circuit plate of a different configuration. The initiator module **250** has a first member **252** that accommodates a central plug **260** that holds a detonator (not shown), similar to the central plug **160** of FIG. **1C**. In this case, the central plug **260** is a separate member that is attached to the first member **252** by a fastener **262**, in this case, a screw.

(46) The initiator module **250** also has a second member **254**, which contains a circuit plate **256**, held to the second member **254** by circuit plate clips **258**. Outer clips **270** of the second member **254** engage with recesses **272** of the first member **252** to releasably attach the first and second members **252** and **254** together.

(47) In this case, the initiator module **250** is configured to accommodate wires to connect the detonator to contacts located at a periphery of the circuit plate **256**. Here, the first member **252** has a central opening **274** to allow wires from the detonator to pass through the central opening **274** into a passage **276** of the first member **252**. The central opening **274** has a flared entrance **275** to simplify feeding wires of the detonator through the central opening **274**. The passage **276** includes a curved guide **278** that extends from the central opening **274** toward the second member **254** and radially outward away from the central opening **274**, along a curved path, toward the periphery of the first member **252**. The passage **276**, with the guide **278**, steers detonator wires to the periphery of the first member **252** to simplify connection of the wires to contacts at the periphery of the circuit plate **256**.

(48) FIG. **3A** is an isometric view of a frame **300** according to another embodiment. The frame **300** can be used as one of the frames **164** in the charge module **132** of the perforation tool **100** of FIGS. **1A-1D**. The frame **300** has a generally cylindrical shape with a central axis **302**. In this case, the frame **300** is a single unit with a first end **304** and a second end **306** opposite from the first end **304**. The central axis **302** extends from the first end **304** to the second end **306**. Each of the first end **304** and the second end **306** have a generally circular shape as the ends of the cylindrical shape of the frame **300**.

(49) A side **308** of the frame **300** connects the first end **304** to the second end **306**. A plurality of recesses **310** are formed in the side **308**. The recesses **310** are generally circular and extend into the side **308** toward the central axis **302** and have diameter that generally declines toward the central axis **302**. Each recess **310** generally defines an outer wall **312** of the recess **310**, which is a surface extending from the side **308** toward the central axis **302**. The outer wall **312** maintains a generally circular cross-section while tapering in dimension toward the central axis **302**. This outer wall **312** will generally take a shape that facilitates placement of charges in the recesses **310** that have shapes

designed to project a jet of ballistic discharge radially outward from the central axis **302**. The recesses **310** are generally arranged in a coplanar arrangement, where a plane defined by the centroids of the three recesses **310** is perpendicular to the central axis **302** of the frame **300**.

(50) Where the outer wall **312** of the recess **310** approaches the first end **304** and second end **306** of the frame **300**, a notch **314** is provided in the side **308** of the frame **300** from the outer wall **312** of the recess **310** to the respective first end **304** or second end **306**. The notch **314** provides a foundation for a restraint **316** that operates to secure a shaped charge in the recess **310**. The restraint **316** extends from a side wall **318** of the notch **314** in a direction generally across the notch **314** toward the opposite side wall of the notch **314**, which faces the side wall **318**. The notch **314**, in this case, has a rectangular profile that extends radially inward from the side **308** of the frame **300** toward the central axis **302**. Each notch **314** thus has two side walls **318**, which are substantially parallel, and a floor **320** connecting the two side walls **318** and perpendicular to both side walls **318**. The floor **320** of each notch **314** is planar here, but could be curved in other embodiments. As noted above, each end of the frame **300**, the first end **304** and the second end **306**, has a notch **314**, and the notches **314** at the two ends **304** and **306** are aligned in a direction parallel to the central axis **302**. The two side walls **318** of each notch **314** are substantially parallel, and substantially parallel to a plane midway between them that runs through the central axis **302**. Thus, the two side walls **318** do not run in a radial direction in this embodiment. Alternately, the two side walls **318** could run in radial directions, in which case the two side walls **318** would not be parallel. The side walls **318** are generally flat, but may be any convenient shape, curved, angled, beveled, and the like.

(51) There are two restraints **316** for each recess **310**, one disposed in each opposing notch **314** of a recess **310**. The restraints **316** of a recess **310** extend from opposite side walls **318**, which means that one of the restraints **316** of a recess **310** extends from a side wall **318** on a first side of a plane bisecting the notch **314**, and the other restraint **316** extends from a side wall **318** on a second side of the plane bisecting the notch **314**. The restraints extend from their respective side walls **318** along the edge of the recess **310**. Each restraint **316** extends substantially parallel to the floor **320** of the notch **314**. The restraints **316** are curved in this case, following the circular perimeter of the recess **310** in opposite directions. The restraints **316** are designed to engage with a groove in the rim of a charge liner. The restraints **316** flex outward as the charge liner is pressed into the recess **310**, and snap into the groove of the charge liner to securely hold the charge in place. It should be noted that the restraints **316** of a recess **310** are located at the same elevation above the floor **320** of their respective notches **314**, but may be at different elevations, for example if two grooves are provided in the charge liner to engage the two restraints **316**.

(52) The frame **300** has a nub **322** that extends from the first end **304** of the frame **300**. The nub **322** is cylindrical here, but may be any convenient shape. For example, the nub **322** can have the shape of any regular polygon, for example square, triangular, pentagonal, and the like. A circular opening **325** at the end of the nub **322** leads to a passage through the frame **300** along the central axis **302**. The passage may be the same as the passage **174** described above in connection with the perforation tool **200**. The passage is not visible in FIG. 3A, but extends through the frame **300** to the second side **306** thereof to provide a conduit for the electrical conductor **176** described above.

(53) Adjacent to the center of the floor **320** of each notch **314** of the frame **300**, extending from the first side **304** and substantially perpendicular thereto, is a first alignment feature **324**. Since there are three notches **314** in the frame **300**, there are three first alignment features **324**. In general, one or a plurality of first alignment features **324** may be provided. The alignment features **324** can have any arrangement on the first end **304**. For example, the alignment features **324** could be located between the notches **314** close to the edge of the first end **304**, or near the nub **322**. Alternately, two alignment features **324** could be used arranged on opposite sides of the nub **322**. Any convenient arrangement of alignment features **324** could be used. The alignment features **324** here are posts that extend perpendicular to the first end **304** and are uniformly distributed around the nub **322**.

These posts have a plus shape in cross-section. Any type of alignment feature **324** could be used. For example, bumps, ridges, and other kinds of protrusions can be used, and can be mixed with recesses and/or notches, and/or posts with different cross-sectional profiles, for example square, hexagonal, or with an irregular shape. The alignment features **324** mate with alignment features on the second end **306** of the frame **300** to maintain alignment of two frames **300**, one to the other. The alignment features **324** can also be used, if desired, to constrain rotation of one or more frames **300** within a perforation tool such as the perforation tool **100**.

(54) Where cylindrical connectivity features are used to connect a first frame with a second frame, as described herein, the first and second frame are operable in any rotational alignment. The axial features of the frames maintain electrical and fluid communication between the first frame and the second frame, so that the frames can be continuously rotated, one with respect to the other, and operability of the frames maintained. Where a discharge of a shaped charge creates a torque, the frames may relatively rotate following a discharge, potentially changing the direction of subsequent discharges. If discharge is intended in certain directions, rotation of a frame following a discharge can send subsequent discharges in unintended azimuthal directions. The alignment features described herein can be used to constrain unintended rotation of the frames, one with respect to another.

(55) FIG. 3B is a cross-sectional view of a restraint **316** of FIG. 3A. The restraint **316** has a generally irregular shape in cross-section, with a first end **340** that has a rectangular shape and a second end **342** opposite from the first end **340** that has an irregular shape. The restraint **316** has a first surface **344** and a second surface **346** perpendicular to the first surface **344**. The first surface **344** has a first width, and the second surface **346** has a second width less than the first width. The restraint **316** has a third surface **348** also perpendicular to the second surface **346**, and therefore parallel to the first surface **344**, with the second surface **346** connecting the first surface **344** and the third surface **348**. The third surface **348** has a third width less than the first width. The restraint **316** has a fourth surface **350** perpendicular to the first surface **344**, and therefore parallel to the second surface **346**. The fourth surface has a fourth width that is less than the second width. The restraint **316** has a fifth surface **352** that connects the third surface **348** to the fourth surface **350**. The fifth surface **352** has a straight portion **354** connected to the third surface **348** and a curved portion **356** connected to the fourth surface **350**. The straight portion **354** forms an angle with the third surface **348** that depends on the length of the third surface **348**, and may be between about 120° and about 170°, for example about 150°. The curved portion **356** may be circular, or otherwise rounded in a specific, or non-specific way. Here, the curved portion **356** has a radius of curvature that is about the same as the fourth length. The radius of curvature of the curved portion **356** depends on the dimensions of the other surfaces. The cross-sectional profile of the restraint **316** is selected to mate with a groove in the charge cup that holds the shaped charge, and to have a shape that successfully secures the restraint **316** in the groove.

(56) Referring again to FIG. 3A, each restraint **316** extends from a base **317** attached to the side wall **318** of the notch **314**. The base **317** has curved surfaces that connect the various surfaces of the restraint **316** to the side wall **318**. Each curved surface of the base **317** connects from the side wall to one of the surfaces of the restraint **316** described above. The base **317** thus has a dimension that matches a corresponding dimension of the restraint **316** where the base **317** meets the restraint **316**, and increases in areas of the base **317** approaching the side wall **318**. The base **317** is thus thicker than the restraint **316** to provide a strong foundation for attaching the restraint to the side wall **318**.

(57) Each restraint **316** has a latch feature **329** at a distal end of the restraint **316**. In this case, the latch feature **329** has the shape of a triangular prism with two parallel right triangular surfaces connected by rectangular surfaces. The right angle of the latch feature **329** abuts the first surface **344** of the restraint **316** described above in connection with FIG. 3B. The latch feature **329** has a thickness, distance of separation of the triangular surfaces, that is less than the first width of the

first surface **344**. In this case, the thickness of the latch feature **329** is about half the first width, but can be more or less than half the first width in other embodiments. The right angle of the latch feature **329** is positioned near and end **331** of the restraint **316**, and the latch feature **329** rises from the right angle to an edge **333**. A surface of the latch feature **329** slopes downward from the edge **333** along the restraint **316**, such that the tallest part of the latch feature **329**, having the edge **333**, is near the end **331** of the restraint **316**, and the latch feature **329** narrows in a direction along the restraint **316** away from the end **331**.

(58) FIG. 3C is an isometric view of the frame **300** looking toward the second end **306**. The second end **306** has an opening **326** that fluidly communicates with the nub **322** of FIG. 3A on the first end **304** by the passage extending through the frame **300**. The opening **326** has a diameter larger than the diameter of the passage, as described in connection with the perforation tool **100**. The diameter of the opening **326** is substantially the same as the outer diameter of the nub **322**, so the nub **322** can fit into the opening **326** to stack multiple frames **300** together. The nub **322** can also fit with other components of a perforation tool, as described above. The opening **326** also accommodates coupling of other components of a perforation tool.

(59) A plurality of second alignment features **328** are formed in the second end **306** of the frame **300**. These alignment features **328** can engage with the first alignment features **324** of the first end **304**. There are six second alignment features **328** on the second end **306**, where there are only three alignment features **324** on the first end **304**. As above, in general, one or a plurality of second alignment features may be provided. The six alignment features **328** here are recesses shaped to match the posts at the first end **304** of the frame **300** to allow two frames **300** to be engaged in two different relative orientations. The recesses **310** of two adjacent frames **300** coupled together can be aligned, pointing in the same direction, or the recesses **310** can be azimuthally displaced by 30°. The relative orientation of adjacent frames can be selected in this way by providing alignment features having any desired relationship to allow different relative orientations. It should be noted that the spacing of alignment features can be any useful spacing to allow positioning adjacent frames with a variety of different angular displacements.

(60) Each recess **310** of each frame **300** also features a finger notch **330**. The finger notch **330** is formed in the side **308** of the frame **300**. The finger notch **330** is formed in the side **308** between the first and second ends **304** and **306** of the frame, on only one side of the recess **310**. The finger notch **330** extends through the outer wall **312** of the recess **310**, providing access to the recess **310** from the side. The finger notch **330** simplifies the process of extracting charges, for example spent charges, from the frame **300** by providing a lateral extraction method that minimizes stress on the restraints **316** as the charge is removed from the frame **300**. The finger notch **330** can have any convenient shape. Here, the notch **330** has a flat floor **332** and substantially straight side walls **334**. The side walls **334** are not parallel here, but there is no reason they could not be parallel.

(61) FIG. 3D is an isometric view of a frame **380** according to another embodiment. This isometric view is from the same perspective as the view of FIG. 3C. The frame in FIG. 3D is a variant that has only two of the recesses **310**. The two recesses **310** of the frame **380** are also arranged in the same coplanar fashion as the recesses **310** of the frame **300** in FIG. 3C. Here, two recesses **310** are arranged at an angle α that is from 120° to 180°. Such frames **380** can be made with different angles α such that charge geometry for a single perforation tool can be distributed among a plurality of such angles and then selected using any suitable firing method. A single perforation tool can be fitted with an assortment of frames **300** and frames **380** to provide flexibility regarding the directions chosen for perforation.

(62) While the foregoing is directed to embodiments of the present invention, other and further embodiments of the present disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A perforation tool, comprising: a housing; a plurality of frames disposed within the housing, wherein each frame of the plurality of frames has a central axis and a plurality of liners, wherein each frame of the plurality of frames is axially aligned, and wherein each liner of the plurality of liners is configured to secure a respective shaped charge, wherein each frame of the plurality of frames comprises a central passage and an electrical conductor disposed in the central passage; a capsule configured to provide an ignition source for the respective shaped charge, wherein the capsule is disposed within the electrical conductor; an initiator module disposed in the housing; and a bulkhead member disposed in the housing and forming a seal with the housing, wherein the bulkhead member is disposed in the housing separating the initiator module from each frame of the plurality of frames within the housing.
2. The perforation tool of claim 1, wherein the bulkhead member electrically isolates the initiator module from the plurality of shaped charges.
3. The perforation tool of claim 1, wherein each liner of the plurality of liners has an axis transverse to the central axis.
4. The perforation tool of claim 1, wherein the housing comprises one or more perforation grooves having a first thickness less than a second thickness associated with the housing.
5. The perforation tool of claim 1, wherein a first liner of the plurality of liners is configured to orient a first shaped charge in a first direction, and wherein a second liner of the plurality of liners is configured to orient a second shaped charge in a second direction different than the first direction.
6. The perforation tool of claim 1, comprising one or more wireless shaped charges.
7. A perforation tool, comprising: a housing; one or more wireless shaped charges; a plurality of frames disposed within the housing, wherein each frame of the plurality of frames has a central axis and a plurality of liners, wherein each frame of the plurality of frames is axially aligned, and wherein each liner of the plurality of liners is configured to secure a respective wireless shaped charge of the one or more wireless shaped charges, wherein each frame of the plurality of frames comprises a central passage and an electrical conductor disposed in the central passage; a capsule configured to provide an ignition source for the one or more wireless shaped charges, wherein the capsule is disposed within the electrical conductor; an initiator module disposed in the housing; and a bulkhead member disposed in the housing and forming a seal with the housing, wherein the bulkhead member is disposed in the housing separating the initiator module from each frame of the plurality of frames within the housing.
8. The perforation tool of claim 7, wherein the bulkhead member has an additional central passage, and an additional electrical conductor disposed in the additional central passage.
9. The perforation tool of claim 8, wherein the bulkhead member has an insulating member disposed in the additional central passage, and wherein the additional electrical conductor disposed in the additional central passage is within the insulating member.
10. The perforation tool of claim 7, wherein an outer surface of the electrical conductor contacts an inner surface of the central passage.
11. The perforation tool of claim 7, wherein a first frame of the plurality of frames has a first central passage, a second frame of the plurality of frames has a second central passage, and the second frame is ballistically coupled to the initiator module via the second central passage.
12. The perforation tool of claim 7, wherein a first liner of the plurality of liners is configured to orient a first shaped charge in a first direction, and wherein a second liner of the plurality of liners is configured to orient a second shaped charge in a second direction different than the first direction.
13. A perforation tool, comprising: a housing; a plurality of frames disposed within the housing,

wherein each frame of the plurality of frames has a central axis and a plurality of liners, wherein each frame of the plurality of frames is axially aligned, wherein a first liner of the plurality of liners is configured to orient a first shaped charge in a first direction, and wherein a second liner of the plurality of liners is configured to orient a second shaped charge in a second direction different than the first direction, wherein each frame of the plurality of frames comprises a central passage and an electrical conductor disposed in the central passage; a first capsule configured to provide a first ignition source for the first shaped charge, wherein the first capsule is disposed within the electrical conductor; an initiator module disposed in the housing; and a bulkhead member disposed in the housing and forming a seal with the housing, wherein the bulkhead member is disposed in the housing separating the initiator module from each frame of the plurality of frames within the housing.

14. The perforation tool of claim 13, wherein the first direction is in a first azimuthal direction, and the second direction is in a second azimuthal direction.

15. The perforation tool of claim 13, wherein each frame of the plurality of frames is configured to rotate within the housing.

16. The perforation tool of claim 13, wherein each frame of the plurality of frames is configured to be independently rotatable within the housing.

17. The perforation tool of claim 13, wherein the first direction is opposite to the second direction.

18. The perforation tool of claim 13, comprising one or more wireless shaped charges.

19. The perforation tool of claim 13, wherein the first capsule is configured to provide the first ignition source for the second shaped charge.

20. The perforation tool of claim 19, comprising a second capsule configured to provide a second ignition source for a third shaped charge upon detonation of the first capsule, wherein the second capsule is disposed within the electrical conductor.
